

Transport Layer Protocols (TCP) Examination Lab

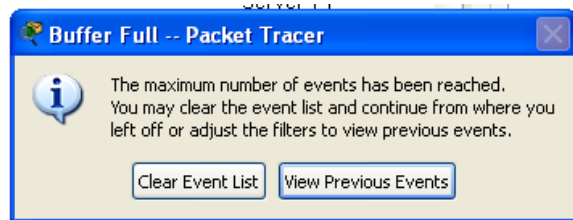
Objectives:

Capture traffic and observe the PDUS for TCP when a HTTP request is made.

Task 1: Observe TCP traffic exchange between a client and server.

Step 1 – Run the simulation and capture the traffic.

- Enter **Simulation** mode.
- Check that your Event List Filters shows only **HTTP** and **TCP**.
- Click on the PC1. Open the **Web Browser** from the **Desktop**.
- Enter **www.bracu.ac.bd** into the browser. Clicking on **Go** will initiate a web server request. Minimize the Web Client configuration window.
- A TCP packet appears in the **Event List**, as we will only focus on TCP the DNS and ARP packets are not shown.
- Click the **Auto Capture / Play** button to run the simulation and capture events.
- Sit tight and observe the packets flowing through the network.



- When the above message appears Click “View Previous Events”.
- Click on PC1. The web browser displays a web page appears.

Step 2 – Examine the following captured traffic.

Our objective in this lab is only to observe TCP traffic.

	Last Device	At Device	Type
1.	PC1	Switch 0	TCP
2.	Local Web Server	Switch 1	TCP
3.	PC1	Switch 0	HTTP
4.	Local Web Server	Switch 1	HTTP
5.	PC1 (after HTTP response)	Switch 0	TCP
6.	Local Web Server	Switch 1	TCP
7.	PC1	Switch 0	TCP

- As before find the following packets given in the table above in the **Event List**, and click on the colored square in the **Info** column.
- When you click on the Info square for a packet in the event list the **PDU Information** window opens. If you click on these layers, the algorithm used by the device (in this case, the PC) is displayed. View what is going on at each layer.

For packet 1::

Click onto "Inbound PDU details" tab. Scroll down and observe the TCP header.

A. What is this TCP segment created by PC1 for? How do you know what is it for?

Created to establish connection and synchronizes sequence.

I know this by looking at the TCP flag

B. What control flags are visible?

FLAGS:0b00000010 (SYN)

C. What are the sequence and acknowledgement numbers?

SEQUENCE NUMBER:0 (used to track the order of transmitted data segments)

ACKNOWLEDGEMENT NUMBER:0 (used to acknowledge the received data segment)

For packet 2:

Click onto "Inbound PDU details" tab. Scroll down and observe the TCP header.

A. Why is this TCP segment created by the Local Web Server?

Created to synchronizes sequence and acknowledgment.

B. What control flags are visible?

FLAGS:0b00010010 (ACK and SYN)

C. Why is the acknowledgement number "1"?

It means that it has received the data segments up to that point.

For packet 3:

This HTTP PDU is actually the third packet of the "Three Way Handshake" process, along with the HTTP request.

A. Explain why control flags **ACK(Acknowledgement)** and **PSH (Push)** are visible in the TCP header?

ACK is visible because it was acknowledged that the there is a connection.

And PSH is visible there is a HTTP request that needs to send.

For packet 5:

After PC1 receives the HTTP response from the Local Web Server, it again sends a TCP packet to the Local Web server why?

After receiving the HTTP response, it again sends TCP packet to close to connection.

Click onto "Inbound PDU details" tab. Scroll down and observe the TCP header.

A. What control flags are visible?

FLAGS:0b00010001 (ACK and FIN)

B. Why the sequence number is 104 and acknowledge number 254? Note this packet is created after PC1 receives the HTTP response from the server.

It means PC1 has received data up to a sequence number and the data segment number 104 is yet to come.

And acknowledgment number 254 means it is ready to receive further data starting from the sequence number.

For packet 6:

Click onto "Inbound PDU details" tab. Scroll down and observe the TCP header.

What is this packet sent from the webserver to PC1 for?

The TCP packet is sent to close to connection.

What control flags are visible?

FLAGS:0b00010001 (ACK and FIN)

Why the sequence number is 254?

It means PC1 has received data up to a sequence number and the data segment number 254 is yet to receive by the Local Web Server.
